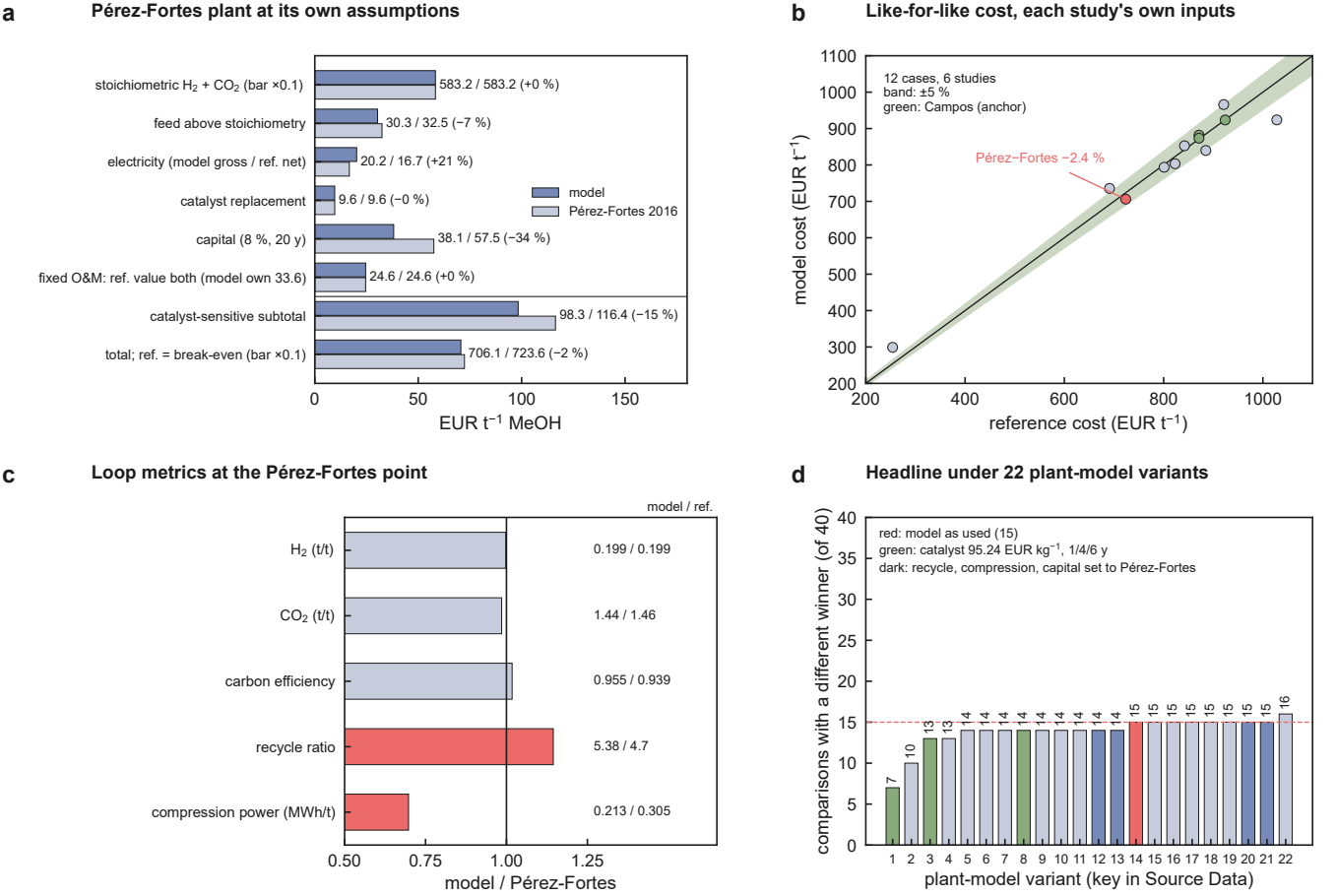

a
Pérez-Fortes plant at its own assumptions
stoichiometric H2 + CO2 (bar ×0.1)
583.2 / 583.2 (+0 %)
feed above stoichiometry
30.3 / 32.5 (−7 %)
electricity (model gross / ref. net)
20.2 / 16.7 (+21 %)
catalyst replacement
9.6 / 9.6 (−0 %)
capital (8 %, 20 y)
38.1 / 57.5 (−34 %)
fixed O&M: ref. value both (model own 33.6)
24.6 / 24.6 (+0 %)
catalyst-sensitive subtotal
98.3 / 116.4 (−15 %)
total; ref. = break-even (bar ×0.1)
706.1 / 723.6 (−2 %)
model
Pérez-Fortes 2016
0
50
100
150
EUR t−1 MeOH
b
Like-for-like cost, each study's own inputs
12 cases, 6 studies
band: ±5 %
green: Campos (anchor)
Pérez–Fortes −2.4 %
model cost (EUR t−1)
reference cost (EUR t−1)
200
300
400
500
600
700
800
900
1000
1100
c
Loop metrics at the Pérez-Fortes point
model / ref.
H2 (t/t)
0.199 / 0.199
CO2 (t/t)
1.44 / 1.46
carbon efficiency
0.955 / 0.939
recycle ratio
5.38 / 4.7
compression power (MWh/t)
0.213 / 0.305
0.50
0.75
1.00
1.25
model / Pérez-Fortes
d
Headline under 22 plant-model variants
red: model as used (15)
green: catalyst 95.24 EUR kg−1, 1/4/6 y
dark: recycle, compression, capital set to Pérez-Fortes
comparisons with a different winner (of 40)
plant-model variant (key in Source Data)
7
10
13
13
14
14
14
14
14
14
14
14
15
15
15
15
15
15
15
16